

Andy Chen

Karl Tizon A0X4351

# Lab 1: Interfacing with LED Using GPIO

## 3. Lab Exercise - Polling (fill in the blanks)

AHB2ENR Mask (Macro Name) = `RCC_AHB2ENR_GPIOAEN`, `RCC_AHB2ENR_GPIOCEN`

AHB2ENR Masking Operation:

- `RCC->AHB2ENR |= RCC_AHB2ENR_GPIOAEN`
- `RCC->AHB2ENR |= RCC_AHB2ENR_GPIOCEN`

GPIO Moder Mask (Macro Names) = `GPIO_MODER_MODE5`, `GPIO_MODER_MODE5_0`

GPIO A MODER Masking Operation:

- `GPIOA->MODER &= ~GPIO_MODER_MODE5`
- `GPIOA->MODER |= GPIO_MODER_MODE5_0`

GPIO A OTYPER Mask (Macro Name) = `GPIO_OTYPER_OT_5`

GPIO A OTYPER Masking Operation: `GPIOA->OTYPER &= ~GPIO_OTYPER_OT_5`

GPIO A PUPDR Mask (Macro Name) = `GPIO_PUPDR_PUPD5`

GPIO A PUPDR Masking Operation: `GPIOA->PUPDR &= ~GPIO_PUPDR_PUPD5`

## 5. Turn-In Questions

1. The mask to clear the mode bits of pin PB11 should be

```
GPIOB->MODER &= ~(3 << 11)
```

2. The line of code to set PC7 to output mode (assuming the MODE register has already been cleared) is:

```
GPIOC->MODER |= (1 << 14)
```

3. The line of code used to read input data from PC3 is:

```
(GPIOC->IDR & (1 << 3))
```